

Task 1a-server.py

```
import socket
import threading

HOST = "0.0.0.0"
PORT = 5000

file_lock = threading.Lock()

def get_grade(marks):
    if marks >= 90:
        return "A+", 4.00
    elif marks >= 86:
        return "A", 4.00
    elif marks >= 82:
        return "A-", 3.67
    elif marks >= 78:
        return "B+", 3.33
    elif marks >= 74:
        return "B", 3.00
    elif marks >= 70:
        return "B-", 2.67
    elif marks >= 66:
        return "C+", 2.33
    elif marks >= 62:
        return "C", 2.00
    elif marks >= 58:
        return "C-", 1.67
    elif marks >= 54:
        return "D+", 1.33
    elif marks >= 50:
        return "D", 1.00
    else:
        return "F", 0.00

def send_message(conn, message):
    conn.sendall((message + "\n").encode())

def ask(conn, reader, message):
    send_message(conn, "INPUT:" + message)
    return reader.readline().strip()
```

```

def handle_client(conn, addr):
    reader = conn.makefile("r")

    try:
        send_message(conn, "Welcome to FAST-NUCES Karachi Campus CGPA Calculator!")

        student_id = ask(conn, reader, "Enter Student ID:")
        subjects = int(ask(conn, reader, "Enter number of subjects:"))

        records = []
        total_credit_hours = 0
        total_points = 0

        for i in range(1, subjects + 1):
            credit_hours = int(
                ask(conn, reader, f"Enter credit hours for Subject {i}:")
            )

            marks = float(
                ask(conn, reader, f"Enter marks for Subject {i}:")
            )

            grade, gpa = get_grade(marks)

            total_credit_hours += credit_hours
            total_points += gpa * credit_hours

            records.append(
                (i, credit_hours, marks, grade, gpa)
            )

        if total_credit_hours > 0:
            cgpa = total_points / total_credit_hours
        else:
            cgpa = 0

        send_message(conn, "")
        send_message(conn, "RESULT")

        for subject, credit_hours, marks, grade, gpa in records:
            send_message(
                conn,
                f"Subject {subject}: Marks = {marks:.2f}, Grade = {grade}, GPA = {gpa:.2f}"
            )

```

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    )

    send_message(conn, f"Overall CGPA: {cgpa:.2f}")

    with file_lock:
        with open("cgpa_log.txt", "a") as file:
            file.write(f"Student ID: {student_id}\n")
            file.write(f"Total Credit Hours: {total_credit_hours}\n")

            for subject, credit_hours, marks, grade, gpa in records:
                file.write(
                    f"Subject {subject}: Credit Hours: {credit_hours}, Marks:
{marks:.2f}, Grade: {grade}, GPA: {gpa:.2f}\n"
                )

            file.write(f"Overall CGPA: {cgpa:.2f}\n")

    print(f"Client {student_id} completed from {addr}")

except Exception as e:
    print("Error:", e)

finally:
    reader.close()
    conn.close()

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen()

print(f"Server running on port {PORT}")

while True:
    conn, addr = server.accept()
    print("Connected:", addr)

    thread = threading.Thread(
        target=handle_client,
        args=(conn, addr)
    )

    thread.start()

```

task 1b – client.py

```
import socket

HOST = "127.0.0.1"
PORT = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))

reader = client.makefile("r")

for message in reader:
    message = message.rstrip()

    if message.startswith("INPUT:"):
        answer = input(message[6:] + " ")
        client.sendall((answer + "\n").encode())
    else:
        print(message)

reader.close()
client.close()
```

```
Server running on port 5000
Connected: ('127.0.0.1', 56846)
Connected: ('127.0.0.1', 56859)
Client 25K-0001 completed from ('127.0.0.1', 56846)
Client 24K-0025 completed from ('127.0.0.1', 56859)
```

```
□
```

Welcome to FAST-NUCES Karachi Campus CGPA Calculator!

Enter Student ID: 25K-0001

Enter number of subjects: 3

Enter credit hours for Subject 1: 3

Enter marks for Subject 1: 92

Enter credit hours for Subject 2: 2

Enter marks for Subject 2: 87

Enter credit hours for Subject 3: 1

Enter marks for Subject 3: 15

RESULT

Subject 1: Marks = 92.00, Grade = A+, GPA = 4.00

Subject 2: Marks = 87.00, Grade = A, GPA = 4.00

Subject 3: Marks = 15.00, Grade = F, GPA = 0.00

Overall CGPA: 3.33

Welcome to FAST-NUCES Karachi Campus CGPA Calculator!

Enter Student ID: 24K-0025

Enter number of subjects: 5

Enter credit hours for Subject 1: 3

Enter marks for Subject 1: 50

Enter credit hours for Subject 2: 3

Enter marks for Subject 2: 50

Enter credit hours for Subject 3: 3

Enter marks for Subject 3: 70

Enter credit hours for Subject 4: 4

Enter marks for Subject 4: 75

Enter credit hours for Subject 5: 1

Enter marks for Subject 5: 57

RESULT

Subject 1: Marks = 50.00, Grade = D, GPA = 1.00

Subject 2: Marks = 50.00, Grade = D, GPA = 1.00

Subject 3: Marks = 70.00, Grade = B-, GPA = 2.67

Subject 4: Marks = 75.00, Grade = B, GPA = 3.00

Subject 5: Marks = 57.00, Grade = D+, GPA = 1.33

Overall CGPA: 1.95

```
server.py  cgpa_log.txt  client.py
cgpa_log.txt
1  Student ID: 25K-0001
2  Total Credit Hours: 6
3  Subject 1: Credit Hours: 3, Marks: 92.00, Grade: A+, GPA: 4.00
4  Subject 2: Credit Hours: 2, Marks: 87.00, Grade: A, GPA: 4.00
5  Subject 3: Credit Hours: 1, Marks: 15.00, Grade: F, GPA: 0.00
6  Overall CGPA: 3.33
7
8  Student ID: 24K-0025
9  Total Credit Hours: 14
10 Subject 1: Credit Hours: 3, Marks: 50.00, Grade: D, GPA: 1.00
11 Subject 2: Credit Hours: 3, Marks: 50.00, Grade: D, GPA: 1.00
12 Subject 3: Credit Hours: 3, Marks: 70.00, Grade: B-, GPA: 2.67
13 Subject 4: Credit Hours: 4, Marks: 75.00, Grade: B, GPA: 3.00
14 Subject 5: Credit Hours: 1, Marks: 57.00, Grade: D+, GPA: 1.33
15 Overall CGPA: 1.95
16
17
```

Task 2

1. What happens if two clients connect at exactly the same time and the server is NOT multithreaded?

The server will handle one client first while the second client waits until the first client's processing is completed.

2. Why is TCP preferred over UDP for this application?

TCP is preferred because it provides reliable, ordered and connection-oriented communication, ensuring that student data and CGPA results are delivered correctly without missing packets.

3. What issue can occur when multiple threads write to cgpa_log.txt at the same time, and how did you handle it?

Multiple threads may write at the same time and mix or corrupt the log data. This was handled using a `threading.Lock()` so only one thread can write to `cgpa_log.txt` at a time.